

Step :1 Install SonarQube Server on WSL

- Requirements for this are
 - Must have JDK 17 and above in wsl
 - sudo apt update
 - sudo apt install openjdk-17-y
 - java -version

```
nikunj@DESKTOP-M4S6FPT:~$ java -version
openjdk version "17.0.15" 2025-04-15
OpenJDK Runtime Environment (build 17.0.15+6-Ubuntu-0ubuntu124.04)
OpenJDK 64-Bit Server VM (build 17.0.15+6-Ubuntu-0ubuntu124.04, mixed mode, sharing)
nikunj@DESKTOP-M4S6FPT:~$ |
```

Step:2 Install PostgreSQL

- sudo apt install postgresql postgresql-contrib -y

Now Create SonarQube DB and User:

- sudo -u postgres psql
- CREATE USER sonar WITH PASSWORD 'sonar'
- CREATE DATABASE sonarqube OWNER sonar;

```
postgres=# CREATE USER sonar WITH PASSWORD 'sonar'
CREATE DATABASE sonarqube OWNER sonar
postgres=# |
```

- \q

```
postgres=# \q|
```

Step:3 Install required Dependencies

- sudo apt install unzip wget gnupg2 -y

```
nikunj@DESKTOP-M4S6FPT:~$ sudo apt install unzip wget gnupg2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-28ubuntu4.1).
wget is already the newest version (1.21.4-1ubuntu4.1).
gnupg2 is already the newest version (2.4.4-2ubuntu17.3).
The following package was automatically installed and is no longer required:
  libltdl7
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
nikunj@DESKTOP-M4S6FPT:~$ |
```

- sudo wget <https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.5.1.90531.zip>
- sudo unzip sonarqube-10.5.1.90531.zip
- sudo mv sonarqube-10.5.1.90531.zip sonarqube

- `sudo chown -R $USER:$USER sonarqube`
- `nano /opt/sonarcube/conf/sonar.properties`
(if opt folder is not available create using command “`mkdir opt`”)

```

GNU nano 7.2 /opt/sonarqube/conf/sonar.properties
# IMPORTANT:
# - The embedded H2 database is used by default. It is recommended for tests but not for
#   production use. Supported databases are Oracle, PostgreSQL and Microsoft SQLServer.
# - Changes to database connection URL (sonar.jdbc.url) can affect SonarSource licensed products.

# User credentials.
# Permissions to create tables, indices and triggers must be granted to JDBC user.
# The schema must be created first.
#sonar.jdbc.username=
#sonar.jdbc.password=
sonar.jdbc.username=sonar
sonar.jdbc.password=sonar
sonar.jdbc.url=jdbc:postgresql://localhost/sonarqube
# Required project info
sonar.projectKey=Sonar_Python
sonar.sources=.
sonar.python.version=3.10

#----- Oracle 19c/21c
# The Oracle JDBC driver must be copied into the directory extensions/jdbc-driver/oracle/.
# Only the thin client is supported, and we recommend using the latest Oracle JDBC driver. See
# https://jira.sonarsource.com/browse/SONAR-9758 for more details.
# If you need to set the schema, please refer to http://jira.sonarsource.com/browse/SONAR-5000
#sonar.jdbc.url=jdbc:oracle:thin:@localhost:1521/XE
|
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-C Copy

```

`sonar.jdbc.username=sonar`

`sonar.jdbc.password=sonar`

`sonar.jdbc.url=jdbc:postgresql://localhost/sonarqube`

`# Required project info`

`sonar.projectKey=Sonar_Python`

`sonar.sources=.`

`sonar.python.version=3.10`

Step: 5 Run the sonarqube

- `cd /opt/sonarqube/bin/linux-x86-64`
- `./sonar.sh start`
- `./sonar.sh status`

Step:6 Log in to sonarqube

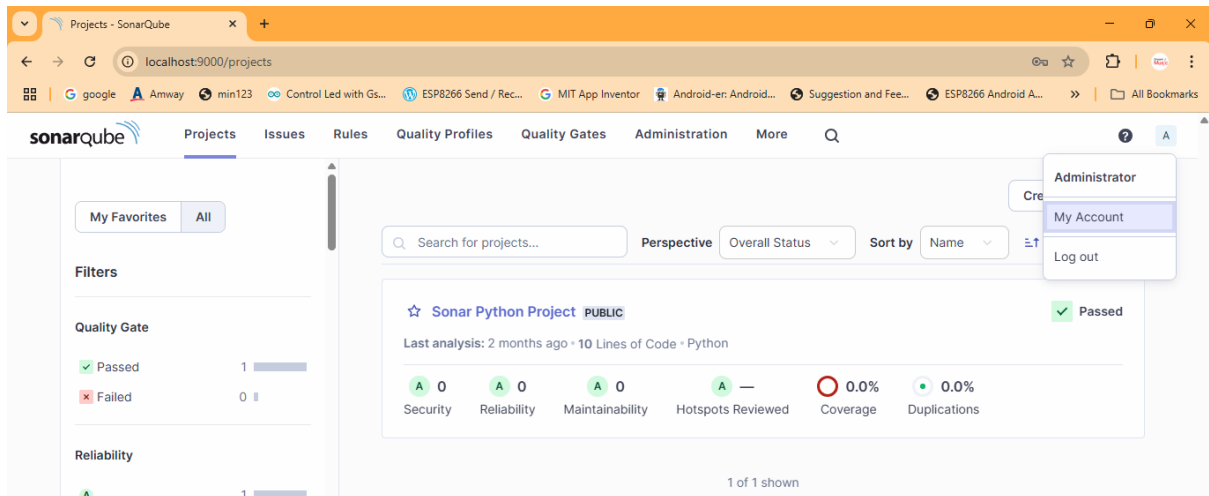
Goto> browser> localhost:9000

(wait for atleast 5-10 min, as it will take a time to start)

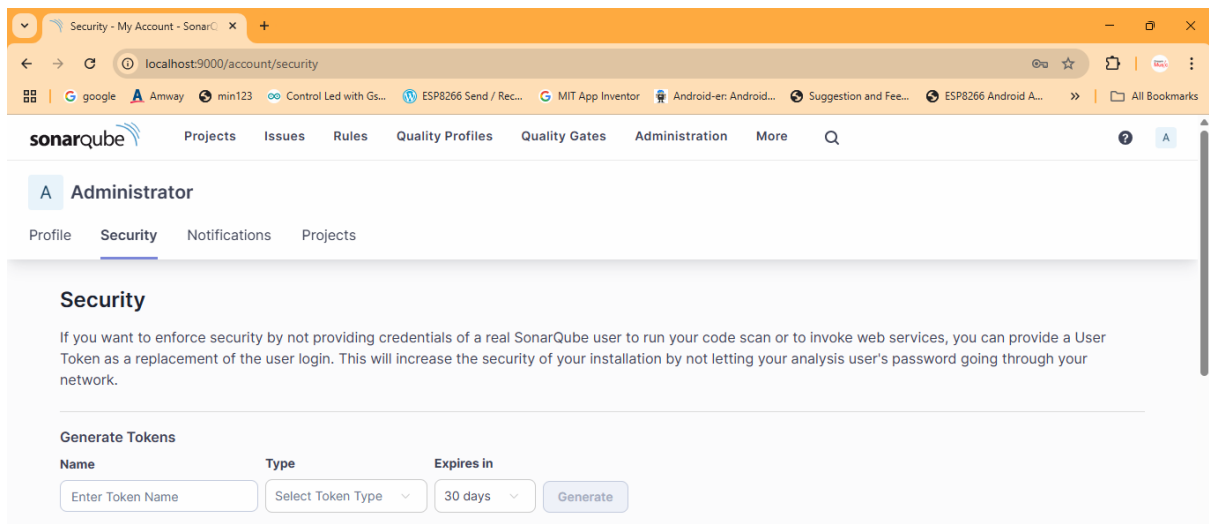
- Default;t username and password are
 - Username: admin
 - Password: admin

- Change the password on first Login

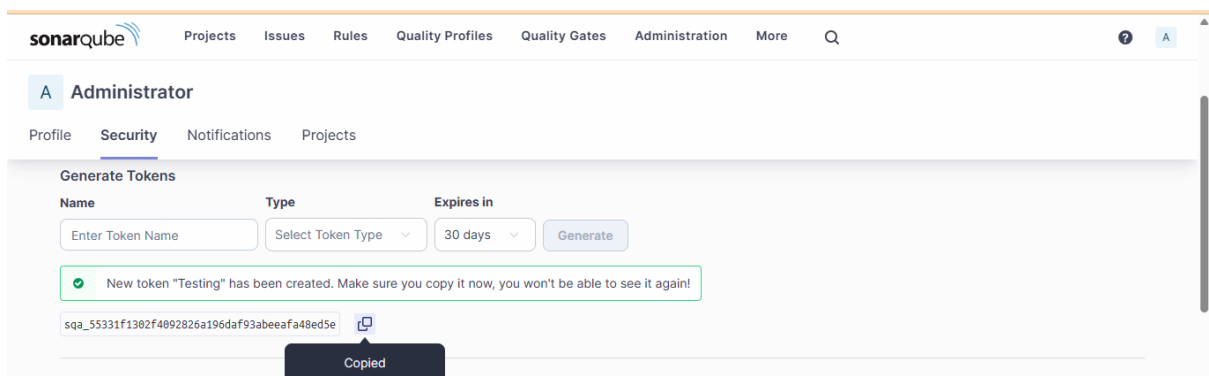
Now click on MyAccount



Click on security tab



Generate New Global Analysis Token



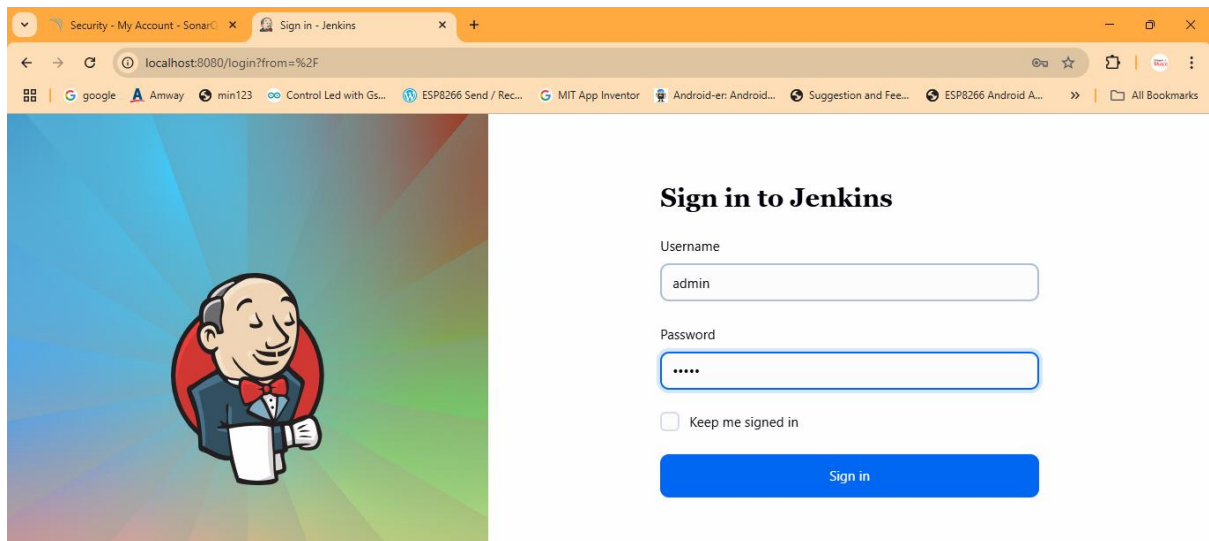
Copy it and save it somewhere securely

JENKINS SONARQUBE INTEGRATION

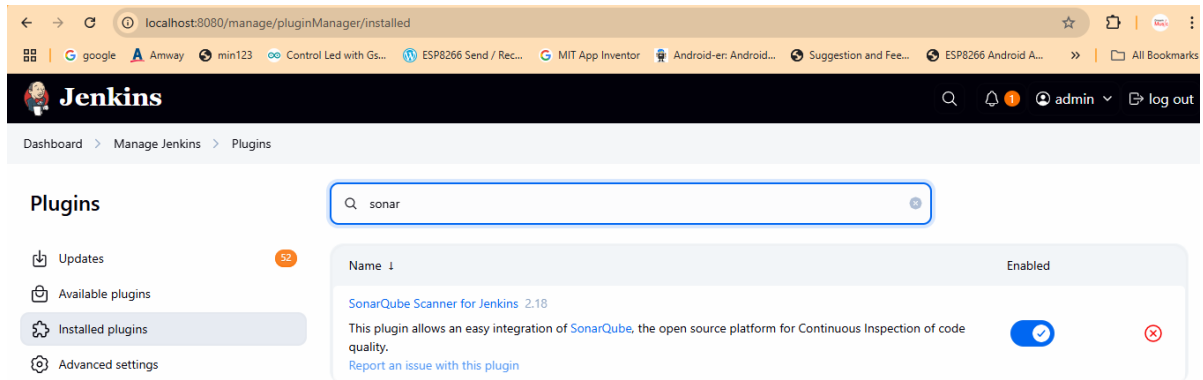
Step:7 start the Jenkins

```
nikunj@DESKTOP-M4S6FPT:/opt/sonarqube/bin/linux-x86-64$ sudo systemctl start jenkins
[sudo] password for nikunj:
nikunj@DESKTOP-M4S6FPT:/opt/sonarqube/bin/linux-x86-64$ |
```

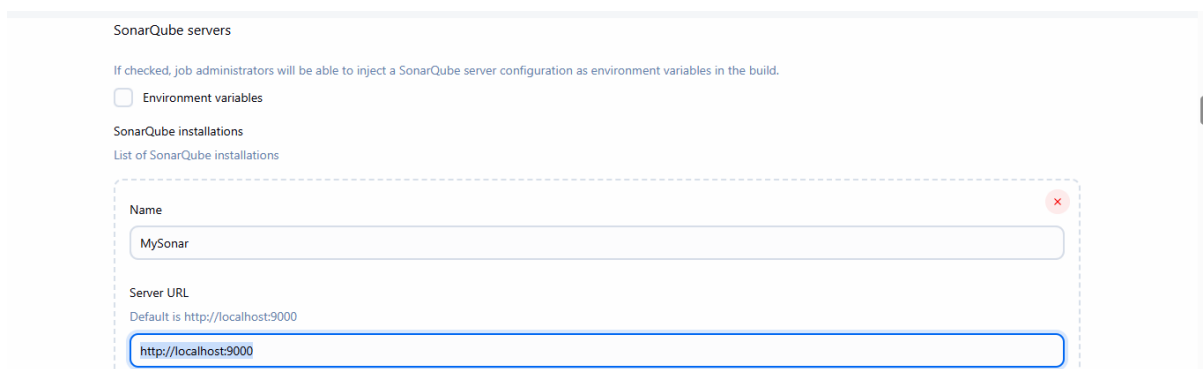
Step: 8 Open Jenkins and add sonarqube to it



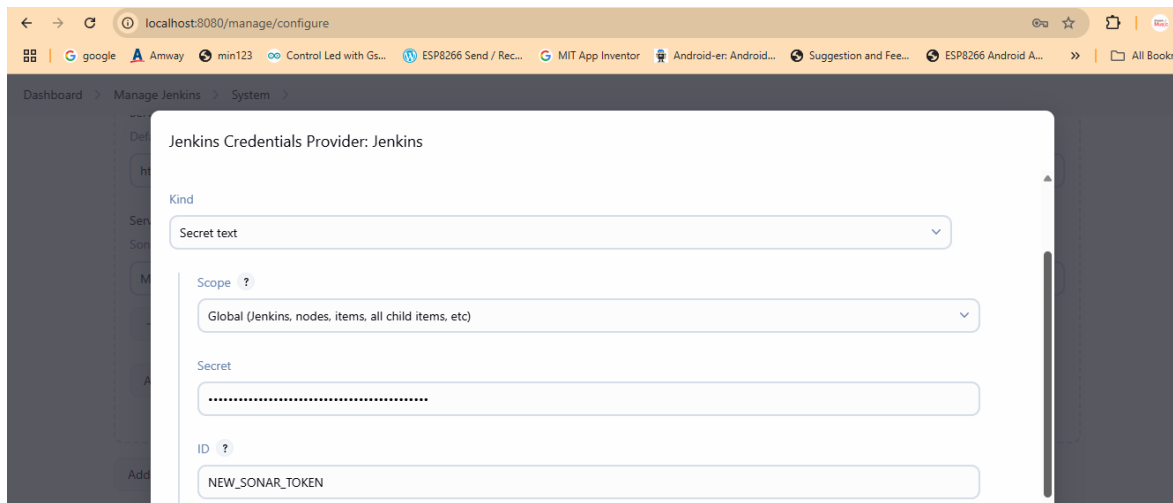
Goto> manage Jenkins> add sonarqube server for jenkins



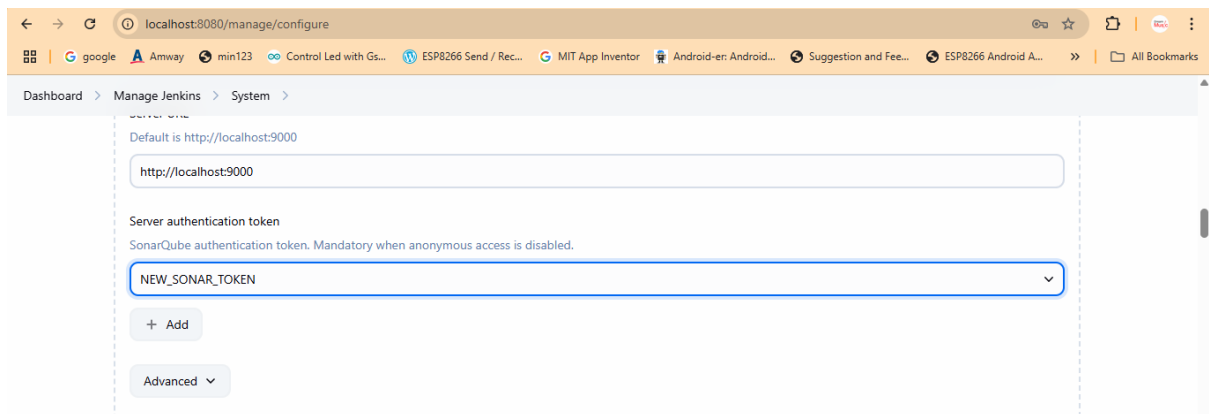
Step:9 goto> manage Jenkins> tools(system)>



Then in Server authentication choose> Jenkins> add new token

A screenshot of the Jenkins web interface showing the 'Jenkins Credentials Provider: Jenkins' configuration page. The page has a sidebar on the left with 'Dashboard', 'Manage Jenkins', and 'System' links. The main content area is titled 'Jenkins Credentials Provider: Jenkins'. It contains four fields: 'Kind' with a dropdown menu set to 'Secret text'; 'Scope' with a dropdown menu set to 'Global (Jenkins, nodes, items, all child items, etc)'; 'Secret' with a text input field containing a series of dots; and 'ID' with a text input field containing 'NEW_SONAR_TOKEN'.

Now in server authentication dropdown > choose newly added token>

A screenshot of the Jenkins web interface showing the 'Server authentication' configuration page. The page has a sidebar on the left with 'Dashboard', 'Manage Jenkins', and 'System' links. The main content area is titled 'Server authentication'. It contains a text input field for 'Default is http://localhost:9000' with the value 'http://localhost:9000'. Below it is a section for 'Server authentication token' with the description 'SonarQube authentication token. Mandatory when anonymous access is disabled.' and a dropdown menu set to 'NEW_SONAR_TOKEN'. At the bottom, there is a '+ Add' button and an 'Advanced' dropdown menu.

Then click on Save

Step:10 Create Jenkins Pipeline> add the pipeline

```
pipeline {
```

```
    agent any
```

```
    environment {
```

```
        SONARQUBE_ENV = 'MySonar'           // Jenkins → Manage Jenkins → Configure System
```

```
        SONAR_TOKEN = credentials('MY_SONAR_TOKEN') // From Jenkins Credentials store
```

```
    }
```

```
    stages {
```

```
        stage('Checkout') {
```

```

    steps {
        git url: 'https://github.com/Nikunj-Java/Sonar_Python.git', branch: 'main'
    }
}

stage('SonarQube Analysis') {
    steps {
        withSonarQubeEnv("${SONARQUBE_ENV}") {
            sh '''
                /opt/sonar-scanner/bin/sonar-scanner \
                -Dsonar.projectKey=Sonar_Python \
                -Dsonar.sources=app \
                -Dsonar.tests=tests \
                -Dsonar.python.version=3.10 \
                -Dsonar.login=${MY_SONAR_TOKEN}
            '''
        }
    }
}

stage('Quality Gate') {
    steps {
        timeout(time: 2, unit: 'MINUTES') {
            waitForQualityGate abortPipeline: true
        }
    }
}
}

```

➤ Build now

➤ check the console output

The screenshot shows the Jenkins Solar Pipeline dashboard. On the left is a sidebar with navigation links: Status, Changes, Build Now, Configure, Delete Pipeline, SonarQube, Stages, Rename, and Pipeline Syntax. The main content area is titled 'Jenkins Solar Pipeline' and features a 'SonarQube Quality Gate' section. This section indicates that the 'Sonar Python Project' has 'Passed' and that 'server-side processing' was a 'Success'. Below this, there is a 'Permalinks' section with a list of links to various build events: 'Last build (#3), 3 min 29 sec ago', 'Last failed build (#1), 2 mo 25 days ago', 'Last unsuccessful build (#3), 3 min 29 sec ago', and 'Last completed build (#3), 3 min 29 sec ago'. At the bottom left, there is a 'Builds' section with a search bar and a refresh icon.

➤ open the Localhost:9000

➤ check the sonarqube projects

The screenshot displays the SonarQube web interface in a browser window at localhost:9000/projects. The interface includes a top navigation bar with links to Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. A left sidebar contains filters for 'My Favorites' and 'All', and a 'Filters' section with 'Quality Gate' (showing 1 Passed and 0 Failed) and 'Reliability' (showing 1 A). The main content area shows a search bar, a 'Perspective' dropdown, an 'Overall Status' dropdown, and a 'Sort by' dropdown. It displays a list of projects, with the first one being 'Sonar Python Project' (PUBLIC), which has a 'Passed' status. Below the project name, it shows 'Last analysis: 4 minutes ago - 10 Lines of Code - Python'. A row of metrics is displayed: Security (A 0), Reliability (A 0), Maintainability (A 0), Hotspots Reviewed (A —), Coverage (0.0%), and Duplications (0.0%). The bottom right corner indicates '1 of 1 shown'.